

Rudra Patel

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Education

Lewis University

May 2026

B.S. Computer Science, Concentration in Artificial Intelligence - GPA 3.9, Dean's List 6x

Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Big Data Systems (Apache Spark, Hadoop, Kafka, Hive)

University of Illinois at Urbana-Champaign (UIUC)

MCS Computer Science - Admitted Fall 2026 - AI/ML for Enterprise Applications

Experience

Software Engineer

May 2025 – May 2026

Hometown Liquor - E Commerce Platform

- Increased monthly revenue by **45% to \$25k** by shipping a full-stack e-commerce platform from scratch with **150+** users, **770+** products using **React**, **TypeScript**, and **FastAPI**
- Secured and scaled backend traffic by designing **92 REST API endpoints** using **FastAPI** with JWT authentication, role-based access control, and rate limiting with audit logging for all requests
- Built an AI-powered search feature that understands natural language queries using **Python**, combining vector similarity search and full-text search in **PostgreSQL** to return relevant results in under **200ms** across 770+ product catalog
- Reduced internal manual inventory checks by **90%** by building a real-time POS sync tool connecting to third party API using async REST and batch processing **100+ products** per batch, eliminating stock-outs across the full catalog
- Implemented CI/CD pipeline via **Docker** containerization enabling consistent, repeatable deployments; wrote **85+** unit and integration tests using **pytest** ensuring reliability across auth, order, and inventory service layers

Research Assistant

Present

Lewis University

- Built a dataset management platform for Lewis Research Lab to centralize, document, and search across all datasets using **React**, and **Go**
- Implemented an AI-powered search engine using **Go**, combining **PostgreSQL** full-text search and semantic vector search, improving search accuracy by **40%** across all lab datasets
- Designed chunked concurrent dataset uploads and constant-memory streaming downloads in Go, handling dataset files up to **50 GB** without timeouts or server memory pressure
- Engineered an upload-time AI pipeline that automatically generates labels, class distributions, and summaries for every dataset within **3–5 seconds**, using a bounded file profiler to validate and clean LLM output before storing metadata

Projects

Distributed LLM Inference Gateway | Go, PostgreSQL, Redis, React, TypeScript, Docker

- Building an open-source, cost-aware LLM gateway in **Go** that standardizes OpenAI, Anthropic, and local Ollama models behind a unified REST API with per-request tracing and structured logging
- Implemented exact-match Redis cache, load tested to handle **9,000+** requests/s, reducing p99 response latency from **2.1s to 15ms**
- Building an async **PostgreSQL** database pipeline to track AI usage, latency, and costs, with an AI-powered caching layer that detects similar user queries to drastically reduce expensive API calls

AI Recipe Service | TypeScript, Java, Spring Boot, AWS, PostgreSQL, Docker, Terraform

[Link](#)

- Built a production AI recipe service in **Java** and **Spring Boot** with a pluggable LLM client for recipes supporting zero-code provider swaps across local Ollama, OpenAI, and Anthropic
- Designed a content-matched AI response cache in **PostgreSQL** with configurable expiration times, cutting redundant AI processing and serving repeat queries in under **10ms** vs 10 to 20s latency
- Added automatic failure handling to the AI generation pipeline including retry logic, rate limiting, and circuit breaking so the app degrades gracefully instead of crashing when the model is slow or unavailable
- Wrote **98+** unit and integration tests, achieving **80%+** test coverage across unit, API, and database layers using real PostgreSQL containers in CI to catch schema regressions before they hit production
- Defined AWS infrastructure in Terraform including **ECS**, **RDS**, **ALB**, **Secrets Manager**, and **S3** with **IAM** roles

Real-Time Event Processor | Node.js, Express.js, TypeScript, Redis, TimescaleDB, Docker

[Link](#)

- Built a distributed event pipeline similar to Kafka processing **2.4M+** events (**28 events/sec**) with zero message loss using **Node.js**, **Express.js**, and **Redis Streams** (consumer groups)
- Reduced dashboard query latency by **55x (1.828s to 33ms)** using **TimescaleDB**, eliminating live computation at query time
- Cut database transactions by **60–70%** via batch processing (**100 events/batch**) with schema validation, and conflict-safe inserts
- Built a REST API using **Express.js** and **Redis** caching with p95 response times of **5–38ms** and 5–10s TTL, serving **2.4M+** records

Hackathons & Competitions

ASA DataFest - American Statistical Association | Python, Pandas, Matplotlib, Seaborn

April 2026

- Built an ETL pipeline in a 48-hour competition using **Python**, cleaning and joining **7.6M+** patient records from global healthcare dataset
- Analyzed **79K** patient records using Python classification algorithms, identifying an **83.6%** hypertension and **74.4%** diabetes disengagement rate, and presented findings and outreach recommendations to competition judges within 48 hours

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, Go, HTML, CSS

Frameworks & Tools: FastAPI, Spring Boot, React, Express.js, Node.js, REST API, Docker, GitHub Actions, Linux, Claude Code

Database & Cloud: PostgreSQL, Redis, Elasticsearch, MySQL, Firebase, AWS (ECS, RDS, S3, CloudWatch, IAM), GCP

Practices & Big Data: AI-Assisted Development, REST APIs, Agile, Microservices, Apache Spark, Hadoop, Kafka, Hive